MILITARY SPECIFICATION

INDICATOR, DIGITAL
(FOR TORPEDO TEST SET MARK 281 MOD 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers a 0 to 9 count digital indicator for use with the Torpedo Test Set Mark 281 Mod 1. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-S-19500/118	Semiconductor Device, Diode, Silicon Types (TX and Non-TX) 1N483B, 1N485B, 1N486B, 1N5194, 1N5195, and 1N5196

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

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MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals

DRAWINGS

Naval Ordnance Systems Command

2639726	Indicator, Digital
2639766	Connector, Digital Indicator

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First articles. Unless otherwise specified in the contract or purchase order, three first articles of the digital indicator shall be manufactured using the methods and procedures proposed for procution. The first articles will be tested as specified in section 4 to determine if the proposed production method will yield an item which meets the technical and design requirements of this specification and the contract or purchase order.

3.2 Materials.

3.2.1 Fungus proof. Materials shall be selected in accordance with MIL-STD-454, requirement 4.

3.2.2 Corrosion resistance. Materials shall be corrosion-resistant or suitably treated to resist corrosion.

3.2.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact.

3.2.4 Fumes. Materials which, under the conditons likely to be encountered in storage or service, liberate deleterious fumes shall not be used.

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3.3 Design and construction. The design and construction shall be in accordance with the requirements specified herein and on NAVORD Drawing 2639726.

3.3.1 Functional schematic. The wiring of the indicator shall conform to figure 1.

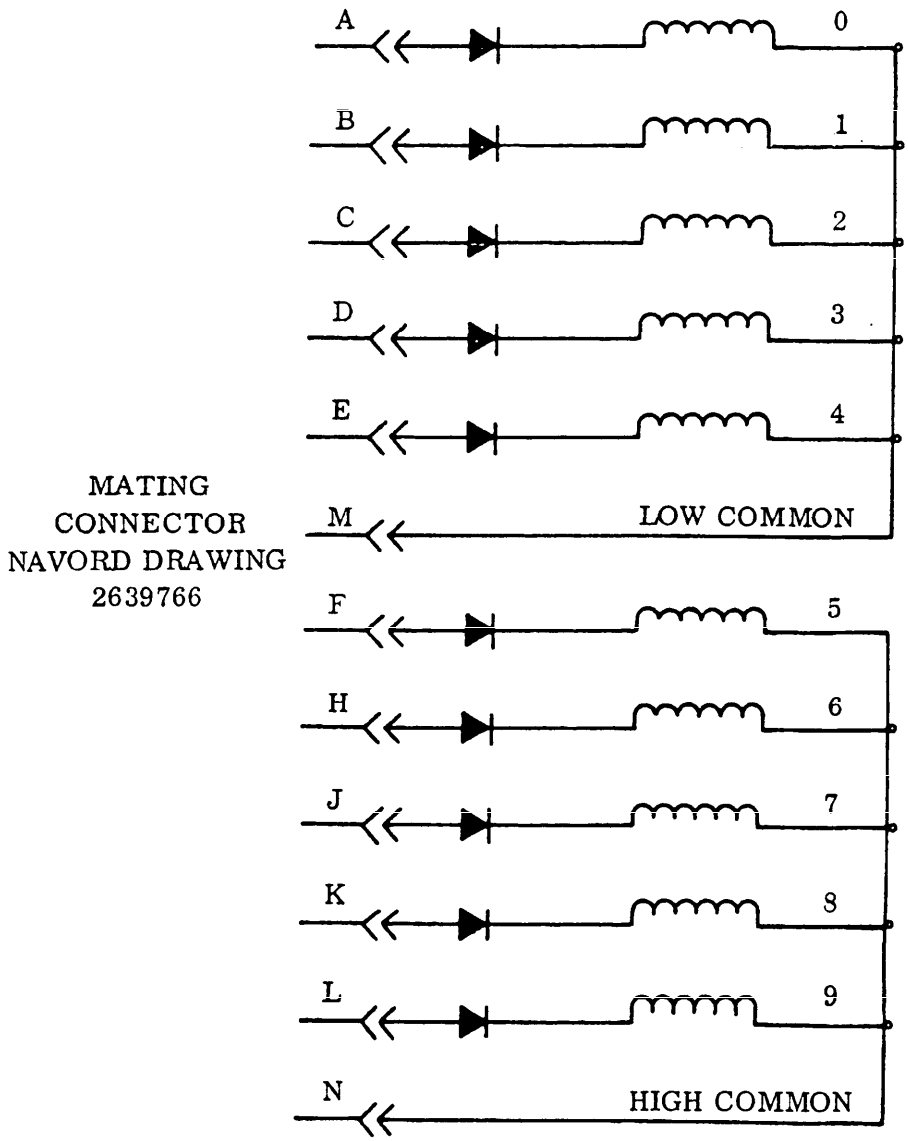


FIGURE 1. DIGITAL INDICATOR, SCHEMATIC

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3.3.2 Series diodes. The diodes to be mounted on the readout shall be type 1N483B or equivalent, in accordance with NAVORD Drawing 2639726 and MIL-S-19500/118.

3.3.3 Common isolation. The resistance between low common and high common shall be greater than 1 megohm.

3.3.4 Coil resistance. The direct-current resistance of each coil shall be 145 ± 14.5 ohms.

3.3.5 Coil inductance. Coil inductance shall not exceed 30 millihenries when measured with a test voltage of 14.0 ± 0.25 volts, 1000 hertz.

3.3.6 Insulation resistance. The resistance between the case and any or all input terminals shall be greater than 25 megohms with 500 volts direct current applied.

3.4 Weight. The digital indicator shall not weigh more than 2.0 ounces.

3.5 Life. The digital indicator should be designed and constructed to withstand a service life of 30,000,000 cycles or 10 years, whichever occurs first. One cycle is a sequence displaying each of the digits once and only once.

3.6 Performance.

3.6.1 Operation. The indicator shall display the selected digit, 0 through 9, when rated voltage is applied. As shown in figure 1, when the ground side of rated voltage is applied to mating connector pin M and the plus side is applied individually to pin A, B, C, D, or E, the numeral 0, 1, 2, 3, or 4, respectively, shall appear. When the ground side of rated voltage is applied to mating connector pin N and the plus side is applied individually to pin F, H, J, K, or L, the numeral 5, 6, 7, 8, or 9, respectively, shall appear.

3.6.2 Rated voltage. The indicator shall display the selected digit when 13 ± 1.3 volts direct current is applied to the proper pins of the mating connector.

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3.6.3 Response time. The indicator shall be capable of changing its display from one digit to any other digit within 0.75 second.

3.6.4 Duty cycle. The indicator shall be capable of continuous operation.

3.6.5 Character position. Characters shall be horizontally centered within ±0.040 inch. The vertical position of each character shall be such that no part of the character is visually cut off by the upper or lower edge of the window, when viewed perpendicular to the indicator face.

3.6.6 Memory. After an indicator has been positioned to a desired character, it shall maintain that character on display in accordance with 3.6.5 after rated voltage is removed. During environmental vibration as specified in 3.7(e), the displayed digit shall remain legible and not become ambiguous.

3.7 Environmental requirements. The indicator shall withstand the following environmental conditions without damage:

(a) High temperature, operating: 73° Celsius
(b) Low temperature, operating: -15° Celsius
(c) High temperature, storage: 85° Celsius
(d) Low temperature, storage: -62° Celsius
(e) Vibration, operating: MIL-STD-167, type I equipment
(f) Vibration, nonoperating: 3.5 g peak at 5 to 500 hertz
(g) Shock, nonoperating: 40 g for 18 milliseconds
(h) Humidity, nonoperating: 95 percent relative at 56° Celsius
(i) Humidity, operating: 85 percent relative humidity at 56° Celsius
(i) Salt fog atmosphere, nonoperating.

3.8 Workmanship. The indicator shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the

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Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the indicator shall be classified as follows:

(a) First article inspection
(b) Quality conformance inspection.

4.2.1 First article inspection. Unless otherwise specified in the contract or purchase order, three indicators shall be manufactured using the methods and procedures proposed for productin. These indicators shall be subjected to the first article inspection procedure of 4.4.

4.2.2 Quality conformance inspection. The procedure of 4.5 shall be performed on each indicator submitted for acceptance under contract (100 percent inspection) to assure that production units meet the requirements of this specification and the contract or purchase order.

4.3 Test conditions. Unless otherwise specified herein, the indicator shall be tested under the following ambient conditions:

(a) Temperature: 25° ± 10° Celsius
(b) Humidity: not exceeding 95 percent relative.

4.3.1 The power supply used to test the indicator shall provide 13 volts direct current adjustable over the range 11.7 to 14.6 volts.

4.4 First article inspection procedure.

4.4.1 First article inspection shall be performed on three first article units in accordance with the test schedule shown in table I.

4.4.2 A failure of any indicator in any test or inspection shall be cause for denial of production approval.

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Table I

TEST SCHEDULE

Test	Requirement	Method	First article unit			Quality conformance
			A	B	C	
Visual and mechanical examination	3.2, 3.3, 3.8	4.6	X	X	X	X
Conditioning	-	4.7.1	X	X	X	X
Performance	3.3, 3.6	4.7.2	X	X	X	X
Rated voltage	3.6.2	4.7.3	X	X	X	X
High temperature, operating	3.7	4.7.4.1	X	X		
Low temperature, operating	3.7	4.7.4.2	X	X		
High temperature, storage	3.7	4.7.4.3	X		X	
Low temperature, storage	3.7	4.7.4.4	X		X	
Vibration, operating	3.7	4.7.4.5	X		X	
Vibration, nonoperating	3.7	4.7.4.6	X		X	
Shock	3.7	4.7.4.7	X	X		
Humidity	3.7	4.7.4.8	X	X		
Salt fog	3.7	4.7.4.9		X		
Fungus	3.2.1	4.7.4.10		X		

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4.5 Quality conformance inspection procedure.

4.5.1 Quality conformance inspection shall be performed on each indicator submitted for acceptance under the contract according to the test schedule shown in table I.

4.5.2 A failure of any indicator in any test or inspection shall be cause for rejection of that indicator.

4.6 Visual and mechanical examination. The indicator shall be examined to verify that the design, construction, physical dimensions, and workmanship are in accordance with this specification and NAVORD Drawing 2639726.

4.7 Test methods.

4.7.1 Conditioning. The indicator shall be cycled through all 10 digits, in accordance with 4.7.2.1, for a total of 12,000 cycles.

4.7.2 Performance.

4.7.2.1 Operation. A voltage of 13 ± 0.1 volts direct current shall be applied to the mating connector pins of the indicator to check conformance to the requirements of 3.6.1, 3.6.5, and 3.6.6. For this test, pins M and N may be connected together.

4.7.2.2 Response time. The time from the initial application of power, specified in 4.7.2.1, until the proper digit is displayed shall be measured with a device accurate to within ±0.05 second.

4.7.2.3 Common isolation. The resistance between pins M and N shall be measured with a device capable of measuring at least 1 megohm with ±5 percent accuracy. The unit shall not be energized during this test.

4.7.2.4 Coil resistance. The direct-current resistance of each coil shall be measured using a device accurate to ±1 percent. The unit shall not be energized during this test.

4.7.2.5 Coil inductance. The inductance of each coil shall be measured using a device accurate to within 5 percent. The voltage applied across the coil for purposes of this test shall be 14.0 ± 0.25 volts alternating current, 1000 hertz.

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4.7.2.6 Insulation resistance. The indicator insulation resistance shall be measured using a device capable of measuring a minimum of 25 megohms with ±10 percent accuracy. For purposes of determining insulation resistance, a test potential of 500 volts direct current shall be applied between the indicator case and each input terminal or all input terminals shorted together.

4.7.3 Rated voltage. The procedure of 4.7.2.1 shall be repeated using the following voltages instead of 13 ± 0.1 volts direct current:

(a) 11.7 ± 0.1 volts direct current
(b) 14.6 ± 0.1 volts direct current.

4.7.4 Environmental tests.

4.7.4.1 High temperature, operating. The indicator, less packaging, shall be subjected to a temperature of 71.1° ± 1.7° Celsius for a period of 3 hours minimum. Within 15 minutes after removal from the temperature chamber, the indicator shall be subjected to the procedure of 4.7.2.

4.7.4.2 Low temperature, operating. The indicator, less packaging, shall be subjected to a temperature of -12.5° ± 2.5° Celsius for a period of 6 hours minimum. Within 15 minutes after removal from the temperature chamber, the indicator shall be subjected to the procedure of 4.7.2

4.7.4.3 High temperature, storage. The indicator, less packaging, shall be subjected to a temperature of 82.5° ± 2.5° Celsius for a period of 24 hours minimum. After returning to room temperature, the indicator shall be subjected to the procedures of 4.6 and 4.7.2.

4.7.4.4 Low temperature, storage. The indicator, less packaging, shall be subjected to a temperature of -58.5° ± 3.5° Celsius for a period of 72 hours minimum. After returning to room temperature, the indicator shall be subjected to the procedures of 4.6 and 4.7.2.

4.7.4.5 Vibration, operating. The indicator, less packaging, shall be subjected to type I mechanical vibration as specified in MIL-STD-167. The indicator shall be energized throughout this test and shall be subjected to the performance test procedures of 4.7.2.1 and 4.7.2.2 while vibrating in each of its three principal axes.

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4.7.4.6 Vibration, nonoperating. The indicator, less packaging, shall be subjected to vibration in accordance with MIL-STD-810, method 514, procedure X, curve AY, for a period of 1 hour in each of the three axes. At completion of this test, the indicator shall be examined for physical damage and shall be subjected to the performance procedures of 4.7.2.1 and 4.7.2.2.

4.7.4.7 Shock. The indicator, less packaging, shall be subjected to shock tests of MIL-STD-810, method 516, procedure I, figure 516.1-2, amplitude b, duration d. At the completion of this test, the indicator shall be examined for physical damage and shall be subjected to the performance procedures of 4.7.2.1 and 4.7.2.2.

4.7.4.8 Humidity. The indicator, less packaging, shall be tested in accordance with MIL-STD-810, method 507, procedure I, except that the upper temperature shall be 52.7° ± 3.5° Celsius instead of 71° Celsius. Within 1 hour, after completion of the test, the indicator shall be subjected to the performance procedures of 4.7.2.1 and 4.7.2.2.

4.7.4.8.1 Humidity operating. If specified by the procuring activity, the indicator shall be subjected to the performance tests of 4.7.2.1 and 4.7.2.2 while at 52.7° ± 3.5° Celsius, 85 percent relative humidity during the last cycle of the humidity test of 4.7.3.8.

4.7.4.9 Salt fog. The indicator shall be tested in accordance with MIL-STD-810, method 509, procedure I. After the test, the indicator shall be examined for evidence of corrosion. After drying and removal of salt deposits, the indicator shall be subjected to the procedures of 4.6 and 4.7.2.

4.7.4.10 Fungus. If specified by the procuring activity, the indicator shall be tested in accordance with MIL-STD-810, method 508, procedure I. After the test, the indicator shall be examined for evidence of deterioration, corrosion, or fungus growth, and the indicator shall be subjected to the performance procedures of 4.7.2.1 and 4.7.2.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified by the procuring activity, the indicator shall be preserved in accordance with MIL-P-116, method II, and packaged and packed in accordance with MIL-STD-794.

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5.2 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Shipment nomenclature for marking shall be, as a minimum, "INDICATOR, DIGITAL, NAVORD Drawing 2639726."

6. NOTES

6.1 Intended use. The indicator is intended for use in Test Set Mark 281 Mod 1. It provides visual digital shaft angle position information when receiving signals from the decoder.

6.2 Ordering data. Procurement documents should specify, but not be restricted to, the following:

(a) Title, number, and date of this specification

(b) Indicator drawing number: NAVORD Drawing 2639726

(c) Quantity ordered should include first article inspection samples that will be expended as a result of first article tests

(d) Information concerning inclusion of optional environmental tests and/or exceptions to this specification or other pertinent documents

(e) Level of packaging and packing desired, if other than as specified in 5.1.

Custodian:
Navy - OS

Preparing activity:
Navy - OS
(Project Number 6695-N005)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-I-82577 (OS), Indicator, Digital (for Torpedo Test Set Mark 281 Mod 1)

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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